FOOD KNOWLEDGE REPRESENTATION

From Food Data to Structured Knowledge

- **Problem:** food, branded-product, nutrient, and measurement data are split across tables, with missing values and distributed relationships.
- **Objective:** transform tabular food data into an interconnected, machine-readable representation for retrieval, validation, integration, and experimentation.

TEAM

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1 · Tabular food data

food.csv, branded_food.csv, food_nutrient.csv, nutrient.csv

2 · Rows become resources

Each row gets a unique URI.

3 · Cells become triples

Each value becomes an RDF triple.

4 · Linked RDF graph

Named graphs link the resources.

Prepared data: food.csv — 20,000 rows · branded_food.csv — 20,000 · food_nutrient.csv — 261,255 · nutrient.csv — 477

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ARCHITECTURE

Our Food Knowledge Representation Pipeline

Starting point: food datasets

1
Data profiling and preparation
Python, Pandas, ydata-profiling

2
RDF conversion and named graphs
Turtle, TriG

3
Ontology modeling
food_ontology.ttl, competency questions

4
Validation
RDF checks, SHACL

5
Semantic mapping and integration
Ontop, mappings

6
Learning over the graph
Graph baseline, GNN, link prediction

7
AI access layer
Questions, retrieval, refusal tests

Knowledge graph (named graphs, ontology, mappings)

Storage and query layer: Apache Jena Fuseki + SPARQL

Logical workflow, not necessarily one automated sequence.

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EVIDENCE CHECK

What We Built and What We Verified

1,200,721

RDF triples loaded into Fuseki, matching the earlier implementation report.

Verified reported result

SERIALIZATIONS

Turtle (.ttl) and **TriG (.trig)**

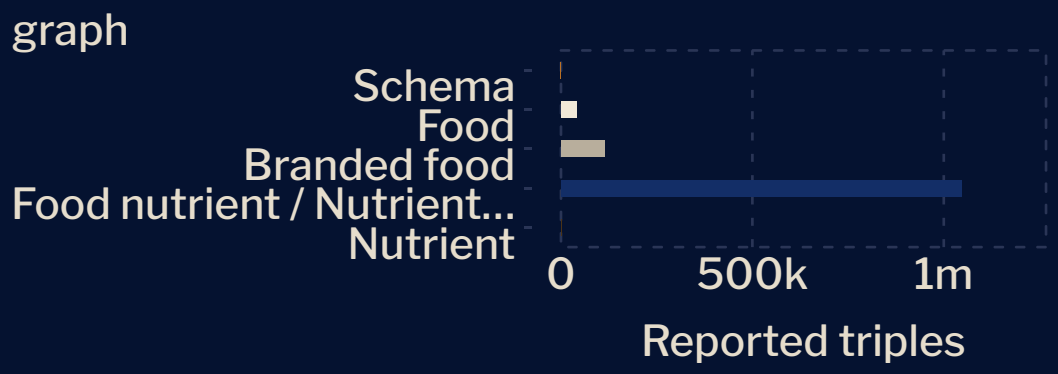
ARTIFACTS

Ontology, competency questions, validation script, broken RDF sample.

EVIDENCE BOUNDARY

Sessions 5–7: components exist in the project; metrics and end-to-end results are not yet verified.

NAMED GRAPH TRIPLE COUNTS



NAMED GRAPH	REPORTED TRIPLES
Schema	38
Food	40,000
Branded food	114,232
Food nutrient / NutrientMeasurement	1,045,020
Nutrient	1,431
Total	**1,200,721**

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RELIABILITY FRAME

Reliability, Design Decisions, and Next Steps

VERIFIED / PRESENT IN PROJECT

- Reported result: 1,200,721 triples in named graphs, loaded to Fuseki
- Components and test artifacts exist in the project
- Design: RDF makes relationships explicit; named graphs organize the data

Contribution: a reproducible path from food tables to a queryable semantic knowledge representation.

STILL TO VERIFY

- Entity-resolution precision and recall
- Leakage-free splits
- Baseline vs GNN metrics
- Access-layer scoring and refusal testing
- CI-gate status

Limitation: end-to-end reliability across integration, learning, and AI access still needs recorded evaluation evidence.

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